

Contents of the shipping box

Nortek order no:

44237-1-620

Type of system:

SIGNATURE 100



Vangkrøken 2
N-1351 RUD
Norway
Teli: +47 6717 4500
Fax: +47 6713 6770
Inquiry@nortek-as.com
www.nortek.no

Instrument type:

- | | | | | |
|---|----------------------------------|--|--|--|
| ☐ Aquadopp | ☐ AWAC | ☐ Vectrino | ☒ Signature100 | ☐ Signature1000 |
| ☐ Aquadopp profiler | ☐ VM AWAC | ☐ Vectrino profiler | ☐ Signature250 | ☐ NortekDVL |
| ☐ Aquadopp DW 3000/6000m | ☐ Vector | ☐ Signature55 | ☐ Signature500 | ☐ NLR DVL |

Software version:

SIGNATURE
DEPLOYMENT

Firmware version:

1.4.5422.2212-11/169

Other:

Cable:

☒ 10m length

☐ Other:

Connector:

☐ 8-pin Inline

☐ 12-pin UW

☒ 6-pin Inline

☐ 7-pin Souriau

☐ Other:

Communication:

☐ RS232

☒ Ethernet

☐ RS422

☐ Other:

Options:

☐ Analog input

☐ Synch

☐ Analog output

☐ Other:

Battery cannister:

☐ Paradopp battery cannister

☐ Single battery aluminium cannister

☐ Double battery aluminium cannister

Battery cables:

☐ 2pin Inline-2pin

☐ 8pin Inline-2pin

☐ 8pin rectangular-2pin

Accessories:

☒ Toolkit

☒ Quick guide

☒ Final test checklist

☐ Seeding material

☐ USB to serial converter RS232

☐ Altronix AL310 USB driver

☐ Recorder kit/ProLog

☐ Battery harness for 2 batteries

Batteries:

☐ Alkaline 50Wh 13.5V

☐ Alkaline 100Wh 13.5V

☐ Alkaline 540Wh 13.5V

☐ Alkaline 540Wh 18V

☐ Alkaline 90Wh 15V(Signature1000)

☐ Alkaline 180Wh 18V(Signature500)

Extra set:

AC/DC Power supply

☐ 15V standard

☐ 48V Signature55
Online

☐ 24V Vectrino

☒ 24V DC/DC &
Signature

☐ EU

☐ UK

☒ US

Other:

Date:

21/6-20

Responsible:

Olaf Pæck

Final test checklist AD2CP



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Order number:
44237-1-620

Name: **Signature 100**
Instrument serial number: **101648**
Frequency: **100 KHz** Main board: **AD2CP-2353**
Firmware versions: **1.4.5422.2212-11/169**

Label checked ☒ OK
Dock test ☒ OK
Baudrate 115200 ☒ OK

Comments:

Tilt check
☒ Pitch up
☒ Roll up
☒ Status bit
☒ Pitch down
☒ Roll down
pitch & roll within +/- 0.2 °

Clock
☒ Set clock
Heading
☒ Up
☒ Down
tolerance: +/- 2 °

Pressure
Psensortemp ☒ OK
tolerance: +/- 0.1 % of
1500 m

Temperature
☒ OK
tolerance: +/- 0.1 °

Beam check

Correct order	Beam Imp	Noise floor	Amplitude in tank	Range
Beam 1 ☒ OK	82 Ω	25 dB	> 80 dB	☒ OK
Beam 2 ☒ OK	82 Ω	25 dB	> 80 dB	☒ OK
Beam 3 ☒ OK	85 Ω	25 dB	> 80 dB	☒ OK
Beam 4 ☒ OK	81 Ω	25 dB	> 80 dB	☒ OK
Beam 5 ☒ OK	543 Ω	25 dB	> 80 dB	☒ OK

Velocity direction

XYZ coordinate system	ENU coordinate system
X ☒ OK	E ☒ OK
Y ☒ OK	N ☒ OK
Z ☒ OK	U ☒ OK
Z ₂ ☒ OK	U ₂ ☒ OK

Head file
☒ Headfile checked
☒ Saved as read only

Serial communication
☐ RS422 **NA**
☐ RS232

Trigger
☐ TTL **NA**
☐ RS485

Recorder erased
☒ OK
Rec size: **256GB**

Ethernet
MAC address: **8C:68:78:00:09:31**
Static IP address:
Set host name: **101648**
DHCP enabled ☒
FTP OK ☒

Licenses

Averaging mode ☒	Wave mode ☐	Vertical velocity ☐	64GB recorder ☐	Calibration license erased ☒
Burst Five beams ☐	Echo Sounder ☒	Dual frequency low ☐	128GB recorder ☐	Production license erased ☒
Bottom track ☐	Ice Measurement ☐	Dual frequency high ☐	256GB recorder ☒	Default configuration set ☒
High Resolution ☐	Altimeter ☐	16GB recorder ☐		

Cable/Harness
Communication ☒ Cable ☒ Communication ☒ Harness
Battery ☒ Battery ☒

Electrical isolation test
50V Ok ☒

External sensors

Power down
☒ OK

Date
Day **29** Month **06** Year **20**

Signature



Tilt and Compass Report

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Details

Instrument Type	Signature100
Instrument S/N	101648
Date	June 11, 2020
Operator	Asle Martinsen
Location	Nortek Factory Norway
Result	Passed

Description

Calibration and verification is performed in a two axis automated jig. Continuous and fixed-point measurements are collected to calibrate and verify the sensor.

Criteria of acceptance for tilt sensor is $\pm 0.2^\circ$.

Criteria of acceptance for compass sensor is $\pm 2^\circ$.

Reference: Digital Protractor Series 950 Pro 3600. Accuracy $\pm 0.05^\circ$.

Tilt Verification Results

Reference ($^\circ$)	Diff. Up		Diff. Down	
	Pitch ($^\circ$)	Roll ($^\circ$)	Pitch ($^\circ$)	Roll ($^\circ$)
-30.00	0.09	-0.05	0.10	0.00
-15.00	0.19	0.08	0.19	0.06
0.00	-0.17	0.01	-0.05	0.06
15.00	-0.07	-0.09	0.09	0.00
30.00	0.19	0.01	0.19	0.08

Compass Verification Results

Reference ($^\circ$)	Heading Diff. Up ($^\circ$)	Heading Diff. Down ($^\circ$)
0.00	0.06	0.23
45.00	0.28	0.28
90.00	0.51	0.06
135.00	0.19	0.20
180.00	0.06	-0.21
225.00	-0.82	0.22
270.00	-0.76	0.19
315.00	-0.75	0.29



Certificate of Calibrations and Tests

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Instrument Information

Customer Reference No.	44237-1-620
Instrument Type	Signature100
Instrument Frequency	100 kHz
Instrument S/N	101648
Head S/N	D-1648
Interface Board S/N	2353
Interface Board Mfr. S/N	4MO0697120013
Digital Board Mfr. S/N	4MO0726280034
Echo Sounder Board Mfr. S/N	4MO0751870001
Analog Board Mfr. S/N	4MO0751860004
Sensor Board Mfr. S/N	4MO0693470069
Interface Board Rev.	H-1
Digital Board Rev.	I-3
Echo Sounder Board Rev.	B-0(High-Power Echo)
Analog Board Rev.	B-0(High-Power 100 kHz)
Sensor Board Rev.	I-0

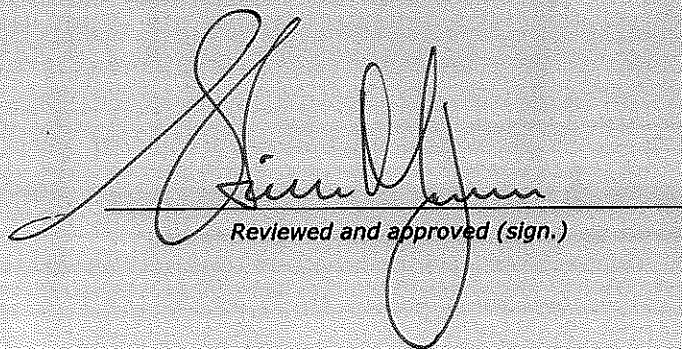
Calibrations and tests performed

Pressure	Passed
Tilt and Compass	Passed

All the tested values are within Nortek AS specifications

June 29, 2020

Date


Reviewed and approved (sign.)



Pressure Report

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Details

Instrument Type	Signature100
Instrument S/N	101648
Pressure Range	1500 dBar
Date	June 24, 2020
Operator	Stian Mysen
Location	Nortek Factory Norway
Result	Passed

Description

Verification is performed in an automated pressure chamber. Fixed-point measurements are collected to verify the sensor.

Criteria of acceptance is $\pm 0.1\%$ of full scale.

Reference: Telemark Technologies - Vessel Pressure Sensor Inlet.

Verification Results

Reference (dBar)	Pressure Diff. (dBar)	Pressure Diff. (% of FS)
31.75	-0.25	-0.02
302.29	-0.32	-0.02
603.22	0.33	0.02
902.70	0.23	0.02
1202.18	-0.33	-0.02
1503.21	-0.07	0.00